

ContinuumOS

Closing the gap between a diagnostic result and a confirmed next step



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THE REAL PROBLEM

The Real Problem Is Not Missing Data

Healthcare teams often have the data. They do not always have accountable closure.

Result available $\neq$ clinician acknowledgement

Referral sent $\neq$ receiving-team response

Message delivered $\neq$ next step confirmed

The failure occurs between systems, teams and decision gates, not inside one record alone.

Problem-framing inputs

Workflow decomposition

Role and authority mapping

Failure-mode analysis

Synthetic episode walkthrough



I Did Not Start by Replacing the EHR

I chose one bounded wedge: **diagnostic closure**.

NOT SELECTED

EHR replacement

SELECTED

Workflow overlay

NOT SELECTED

Generic task tracker

SELECTED

Evidence-gated orchestration

NOT SELECTED

Broad care-orchestration platform

SELECTED

One credible care pathway

NOT SELECTED

Autonomous AI agent

SELECTED

Human-controlled assistance

Narrow scope made the workflow testable, governable and measurable.

My ownership: Problem framing · MVP scope · Operating model · Decision rights · AI controls · Requirements · Prototype · Validation plan

THE CORE RULE

No Milestone Is Inferred from a Neighbouring Milestone

01

Result available

02

Clinician acknowledgement

03

Human direction

04

Handoff

05

Receiving response

06

Confirmed next step



❏ Source evidence establishes context. Authorised people make decisions. Deterministic rules control state progression.

What ContinuumOS Maintains

Source systems remain authoritative. ContinuumOS maintains the orchestration layer.

AUTHORITATIVE SOURCE SYSTEMS

EHR Patient · Encounter	Diagnostics Orders · Results · Reports	Referral System Referral · Receiving response	Communication System Messages · Delivery evidence
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↓ Authorised source reads and references ↓

CONTINUUMOS ORCHESTRATION LAYER

Episode State Current position in pathway	Accountable Owner Assigned role responsible for the next action	Internal Tasks Attributed workflow actions	Exceptions Flags, gaps and escalations
Source References Pointers to authoritative records		Orchestration Audit Full auditable activity log	

Internal workflow actions are recorded here. Clinical source records remain authoritative.

ContinuumOS does not recreate the clinical record. It makes cross-system work visible, owned and recoverable.

Safety changes the workflow

A critical failure mode: stale evidence drives downstream work. *Example: an amended report arrives after acknowledgement.*

Prevent

Bind acknowledgement to report version



Detect

Mark dependent work as stale



Recover

Preserve history and require reassessment

The earlier acknowledgement remains valid historical evidence, not approval for the amended version.

The accepted cost is more visible review and reconciliation. The alternative is silent risk.

Report v1 Acknowledged

Review recorded against report v1

Report v2 Amended

Updated findings issued

Reassessment Required

Require reassessment against v2





AI BOUNDARIES

AI Has a Narrow, Useful Job

AI assists work. It does not make decisions.

Approved assists

→ Source-linked episode summary

Orientation to verified clinical and workflow context

→ Post-approval referral-handoff draft

Operational preparation after human direction

AI cannot

- Diagnose or determine urgency
- Acknowledge a result
- Choose a care pathway
- Independently send a referral
- Accept a referral
- Confirm closure

Approved evidence

Human disposition

Correction captured for evaluation

Source-linked draft

Attributed final action

COMPLETENESS

SOURCE FIDELITY

UNSUPPORTED CONTENT

CORRECTION EFFORT

If AI is unavailable, rejected or stale, the human workflow continues.

Designed for Failure, Not Just the Happy Path

A workflow is only credible if it explains what happens when it breaks.

Designed Exception Paths

Uncertain patient or encounter linkage

Amended report version

Failed action or unavailable service

Incomplete result evidence

Duplicate event

Missing handoff evidence

- ⊗ **Never present failure as success**
- Never create duplicate progression**
- Never advance state without verified evidence**



PROTOTYPE HONESTY

The Prototype Is Deliberately Honest

The prototype demonstrates workflow behaviour, not production readiness.

✓ DEMONSTRATED

❏ NOT REPRESENTED

10 connected screens

Traceable requirements and acceptance criteria

Synthetic JSON representing selected FHIR R4 source resources

Deterministic workflow and exception rules

Pre-written AI outputs

Visible source, audit and failure evidence

Live hospital integrations

Production backend

Live AI model

Source-system write-back

Clinical validation

Launch

Result Review

Handoff

Confirmed Next Step



Workspace

Direction

Receiving Response

Exception handling, source traceability and audit evidence remain visible across the full workflow.

The prototype tests workflow clarity, decision rights, failure handling and AI boundaries before broader implementation claims.

PROPOSED PILOT MEASURES

A Real Pilot Must Measure

PRIMARY OUTCOME

Time from result availability to confirmed next step

WORKFLOW DRIVERS

Episodes with visible owner

Result acknowledgement time

Unresolved handoffs by age

Exception recovery time

Scoped closure rate

AI GUARDRAILS

Source-link coverage

Unsupported-content rate

Human correction rate

Manual fallback rate

These are proposed pilot measures. No targets or improvements are claimed.

From Product Thesis to a Governed Pilot

USER VALUE

Clinic physician

Clear review and decision queue

Care coordinator

Less status chasing; visible exceptions

Referral coordinator

Source-linked handoff and response status

BUYER HYPOTHESIS

Likely sponsor

Network operations · Ambulatory leadership
· Care-coordination leadership

Reason to invest

Reduce unresolved episodes and make
closure measurable

PILOT PROOF REQUIRED

Workflow fit

Data quality

User adoption

Governance approval

Measurable operational value

Scale only after one narrow pathway proves that it reduces invisible work without creating another administrative queue.

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